Unsupervised Traffic Representation and Class Mapping with Self-Organising Maps

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***Abstract*—Network-traffic measurements describe communication patterns, but similar measurements do not necessarily identify the same attack category. The study examined highly imbalanced traffic data containing 10 classes, using a custom batch self-organising map trained without target labels. Clustering grouped the stored representative vectors before training labels mapped the clusters to classes. The frozen confirmatory classifier achieved assessment macro F_1 of 0.324 on folds that kept identical feature vectors together, while geometric diagnostics remained comparable between training and held-out observations. Full class-frequency correction increased rare-class recall but produced thousands of false positives. After the original assessment had been inspected, post-hoc exploration combined finer clustering with partial correction, reaching macro F_1 of 0.458 at 400 clusters. The exploratory gain accompanied greater codebook fragmentation and lower cluster compactness without providing independent confirmation. Nearest-neighbour and classification-tree baselines retained higher macro F_1 of 0.523 and 0.583 on identical folds. The findings showed that useful unsupervised geometry did not ensure reliable attack prediction, while class-frequency correction introduced a substantial precision–recall trade-off.**

I. Introduction

Network-traffic classification distinguishes normal communication from categories of malicious activity. The supplied extract contained 10 unequally represented classes, so evaluation needed to consider rare-class recall and false predictions alongside overall accuracy.

A self-organising map (SOM) represents observations through representative vectors, called codebook vectors, arranged on a two-dimensional grid or lattice. Kohonen introduced the topographic organisation underlying the SOM [7], allowing feature structure to be learned without target labels. However, similarity in traffic measurements does not necessarily imply membership of the same attack category, so representation quality and attack classification require separate evaluation.

The study reused Assignment 2 data for comparison with the selected k-nearest-neighbour (kNN) and classification-tree models. SOM fitting first produced representative vectors from features alone, followed by unsupervised clustering and training-label mapping. The three stages preserved label-free representation while supporting class prediction.

Four research questions guided the analysis, beginning with preprocessing, map capacity and initialisation effects on representation quality. The second concerned codebook clustering and correspondence with the supplied classes, while the third compared the frozen classifier with the Assignment 2 models on identical assessment folds. The fourth examined clustering granularity and class-frequency correction through post-hoc sensitivity analysis.

Development experiments addressed the first three questions before confirmatory assessment, with the procedure fixed before assessment predictions or performance informed selection. The fourth question arose after inspection of the original assessment, although the later exploratory candidates were frozen before further assessment evaluation. The exploratory findings therefore remained separate from the confirmatory estimate. The report presents background and data treatment before experimental design, results and discussion.

II. Background

Representation quality describes feature organisation, whereas classification measures assess target prediction. The background explains the SOM update and diagnostics before introducing clustering, class mapping and evaluation under imbalance.

A. Self-Organising Representation

Let $\mathbf{x}$ denote a transformed observation and m the number of map units, indexed by j. Each unit stores a representative feature vector $\mathbf{w}_j$, called a codebook vector, at a fixed position $\mathbf{r}_j$ on the lattice. The codebook vector has the same features as an observation, while the lattice coordinate locates the unit on the map. The nearest codebook vector identifies the best matching unit (BMU), with the BMU-index function b defined by

$$b(\mathbf{x}) = \arg \min_{j \in \{0,\ldots,m-1\}} \|\mathbf{x} - \mathbf{w}_j\|_2. \tag{1}$$

Feature-space distance associates each observation with a lattice location, with exact ties selecting the lowest unit index.

Neighbourhood coupling links the codebook updates, following the organisation of neighbouring lattice units described by Kohonen [8]. The implementation used a rectangular lattice without wraparound, with epoch index e, source-unit index k and positive neighbourhood radius σ_e. The Gaussian influence $h_{jk}^{(e)}$ from source unit k to destination unit j was

$$h_{jk}^{(e)} = \exp\left(-\frac{\|\mathbf{r}_j - \mathbf{r}_k\|_2^2}{2\sigma_e^2}\right). \tag{2}$$

A broad radius couples distant units, whereas a smaller radius concentrates influence locally. Lattice distance therefore controls neighbourhood updating, while feature distance determines the BMU.

Each batch update first assigned every training observation to a BMU. For source unit k in epoch e, let $n_k^{(e)}$ denote the

assigned count and $\mathbf{a}_k^{(e)}$ the sum of assigned vectors. The destination codebook vector was then updated as

$$\mathbf{w}_j^{\text{new}} = \frac{\sum_{k=0}^{m-1} h_{jk}^{(e)} \mathbf{a}_k^{(e)}}{\sum_{k=0}^{m-1} h_{jk}^{(e)} n_k^{(e)}}. \tag{3}$$

Gaussian weights combined neighbouring BMU aggregates across the complete training partition, without using target labels or an online learning rate. A zero denominator left the previous codebook vector unchanged.

B. Representation Quality

Quantisation error (QE) averages Euclidean distances between observations and the corresponding BMUs. For an evaluated partition X containing n_X observations, the QE function q is

$$q(X) = \frac{1}{n_X} \sum_{\mathbf{x} \in X} \|\mathbf{x} - \mathbf{w}_{b(\mathbf{x})}\|_2. \tag{4}$$

Lower QE indicates closer representative vectors, although absolute errors from different feature transformations have different distance units.

Contextual quantisation error (CQE) expresses map error relative to a single training-centroid representative. Let T denote the training partition and $\boldsymbol{\mu}_T$ the corresponding feature centroid. The CQE function d evaluates partition X as

$$d(X) = \frac{q(X)}{n_X^{-1} \sum_{\mathbf{x} \in X} \|\mathbf{x} - \boldsymbol{\mu}_T\|_2}. \tag{5}$$

For validation, the denominator averaged distances from validation observations to the same training centroid, preserving a training-derived reference. CQE describes relative feature-space distance without measuring classification performance.

Kiviluoto evaluated topology through the first two matching units [6]. Following that approach, topographic error (TE) measures the fraction of observations whose first two distinct BMUs do not share a lattice edge. Stable unit ordering resolves distance ties for both matches, while adjacency requires an edge rather than diagonal contact. Corner units have two edge neighbours, boundary units three and interior units four, so lower TE indicates better agreement between feature proximity and local lattice adjacency.

Occupancy is the fraction of units receiving at least a single BMU assignment, with unassigned units termed dead units. Neighbourhood updates still affect dead codebook vectors. Training occupancy describes codebook use rather than class recognition and therefore needs to be interpreted alongside QE and TE.

C. Codebook Clustering and Class Mapping

Codebook clustering groups the representative vectors stored by the SOM, following the representation-and-clustering approach investigated by Vesanto and Alhoniemi [12]. For K codebook clusters, the K-means algorithm alternates assignment to the nearest centre with updates of the centres. Each observation inherits the BMU cluster, and multiple clusters are allowed to receive the same target class.

Map units represented unequal numbers of training observations, so training-hit weighting gave heavily occupied vectors greater influence on cluster-centre placement. Dead units received zero fitting weight but still obtained cluster assignments. Centre seeding used k-means++ initialisation, proposed by Arthur and Vassilvitskii [1], while clustering consumed only codebook vectors and training hits.

Training labels entered only after clustering to determine class mappings from local counts. For cluster z and class c, let n_{zc} denote the local training count and n_c the total training count of class c. With prevalence-correction strength α, the score function s_{zc} is

$$s_{zc}(\alpha) = \frac{n_{zc}}{n_c^{\alpha}}. \tag{6}$$

The class with the highest score becomes the cluster label, giving ordinary majority mapping at $\alpha = 0$ and full prevalence correction at $\alpha = 1$. Full correction compares fractions of each training class, whereas intermediate values reduce the relative advantage given to rare classes. Frequency correction does not, however, guarantee high predictive precision.

D. Evaluation under Class Imbalance

He and Garcia examined the limitations of aggregate performance under class imbalance [5], motivating evaluation of both precision and recall. For class c, let t_c, f_c and u_c denote true-positive, false-positive and false-negative counts. Precision p_c measures the correctness of predictions for the class, while recall r_c measures the fraction of true class observations recovered:

$$p_c = \frac{t_c}{t_c + f_c}, \qquad r_c = \frac{t_c}{t_c + u_c}. \tag{7}$$

The class score $F_{1,c}$ combines precision and recall, while macro F_1, denoted $F_{1,\text{macro}}$, gives equal weight to every class. For g classes, fixed at 10 in the study, the scores are

$$F_{1,c} = \frac{2 p_c r_c}{p_c + r_c}, \qquad F_{1,\text{macro}} = \frac{1}{g} \sum_{c=0}^{g-1} F_{1,c}. \tag{8}$$

Undefined precision or F_1 is set to zero, keeping classes with no predictions in the macro average. Balanced accuracy averages class recall, ordinary accuracy weights observations equally, and weighted F_1 averages class scores according to true support. Sokolova and Lapalme distinguished the information captured by classification measures [11]. Macro F_1 was primary because rare-class precision mattered alongside recall.

Structural diagnostics complement classification measures without determining classifier selection. Rousseeuw defined silhouette through within-cluster cohesion and nearest-cluster separation [10], with the reported implementation averaging Euclidean scores across all codebook units without hit weighting. Purity measures the observation-weighted fraction belonging to the most frequent class within each cluster, while normalised mutual information (NMI) compares cluster and class information using arithmetic-mean entropy normalisation. The adjusted Rand index (ARI) measures pairwise membership agreement after chance adjustment. Because subdivision increases or

preserves purity, a higher purity alone does not establish better class prediction.

III. Data and Preprocessing

The audit established source identity and feature semantics before distance-based preprocessing and representation experiments. Grouping identical retained inputs then separated repeated vectors across fitting and evaluation partitions.

A. Source and Data Quality

The supplied Assignment 2 extract contained 257,673 observations and 43 descriptive columns, including an identifier, with `attack_cat` as the target. The features and classes corresponded to the University of New South Wales Network Based 2015 (UNSW-NB15) data introduced by Moustafa and Slay [9]. The study retained the supplied extract and repository partitions instead of the original dataset training/testing files.

Table I gives class support across the source and study partitions, retaining the supplied class-identifier order. DoS denotes denial of service. Normal had 93,000 observations, compared with 174 Worms observations. Only seven Worms cases entered development validation, limiting interpretation of precision and recall for that class.

TABLE I
Class support in the frozen study partitions

Class	Source	Training	Validation	Assessment
Normal	93,000	14,880	3,720	74,400
Reconnaissance	13,987	2,239	558	11,190
Backdoor	2,329	372	93	1,864
DoS	16,353	2,617	654	13,082
Exploits	44,525	7,124	1,781	35,620
Analysis	2,677	429	107	2,141
Fuzzers	24,246	3,879	970	19,397
Worms	174	28	7	139
Shellcode	1,511	242	61	1,208
Generic	58,871	9,419	2,355	47,097
Total	257,673	41,229	10,306	206,138

The source contained no missing or non-finite numerical values and no constant features, although service values were missing in 141,321 rows. An explicit `<Missing>` category preserved the affected observations. The binary indicator `is_ftp_login` also contained 26 invalid positive codes of two or four, repaired deterministically to one without deleting rows.

Removing the unique identifier and target exposed 103,989 repeated feature vectors, leaving 153,684 distinct raw groups, including 2,445 with conflicting targets. Conflicting labels limit deterministic separation of identical inputs, but repetition alone did not establish that an observation was invalid. Training, mapping and evaluation therefore retained empirical multiplicity.

B. Distance-Based Feature Treatment

The identifier `id` was excluded because row numbering does not measure traffic similarity, while `stcpb` and `dtcpb` were removed as arbitrary transmission-control-protocol sequence bases. Numerical proximity between sequence identifiers does not imply similar traffic. The audit also verified `tcprtt = synack + ackdat` to floating-point precision, so `tcprtt` was removed to avoid counting both timing components and their sum. Correlated measurements without exact dependence remained.

The retained input comprised 34 numerical columns, two binary indicators and three nominal columns, with numerical parameters and categorical vocabularies fitted on training rows only. Binary indicators retained zero and one without variance scaling, while nominal values used one-hot encoding weighted by $1/\sqrt{2}$. A known-category mismatch therefore contributed a single unit of squared distance, matching a binary mismatch without variance-scaling rare encoded categories.

Standard preprocessing centred numerical columns and divided by the training population standard deviation. Log-standard preprocessing first applied $\log(1+v)$ to non-negative numerical values v in columns with training sample skew of at least two, followed by standardisation. The transformation rule used no labels and clipped no observations, while constant training columns used scale one. Numerical imputation and class resampling were unnecessary.

Both development representations contained 150 coordinates, with log-standard preprocessing transforming 25 columns logarithmically. Unknown nominal levels received an all-zero block, preserving dimension and the training-only vocabulary. Validation contained 197 unknown protocol observations and a single unknown state observation. Under the encoding, unknown-to-known distances were smaller than known-to-known mismatches, and different unseen levels shared the same zero block.

C. Geometry Groups and Partitions

A geometry group contains identical retained inputs after deterministic repair but before fitted transformation, preventing identical vectors from crossing fitting and evaluation partitions. Removing the sequence bases merged a previously distinct pair and reduced the group count from 153,684 to 153,683. The pair crossed historical Assignment 2 assessment folds but remained inside the assessment block.

The original blocks contained 51,535 development rows and 206,138 assessment rows, with stratified group splitting dividing development into 41,229 training and 10,306 validation rows. Assessment contained 122,945 geometry groups, kept intact within five new outer folds. Every partition and fold contained all 10 classes.

Stratification used labels to maintain class coverage, without target-guided feature treatment or SOM updates. The comparison refitted frozen Assignment 2 configurations on the new shared folds, retaining historical scores as a separate protocol reference. Group isolation preserved every observation while evaluating previously unseen feature-vector groups.

IV. Experimental Methodology

The design separated unsupervised representation experiments from classifier selection and locked confirmatory assessment, with later sensitivity studies remaining post-hoc.

A. Implementation and Training

The SOM update and diagnostics were implemented directly with NumPy/SciPy, while scikit-learn supplied preprocessing and K-means clustering. Initial codebooks sampled training rows without replacement, allowing repeated vectors, while five fixed seeds measured initialisation variation. Validation supplied diagnostics without updating codebook vectors.

For map side length ℓ, the 70-epoch schedule reduced radius exponentially from $\ell/2$ to 0.75 during the first 40 epochs, followed by 30 epochs at 0.75. The original schedule had reached minimum radius only at the final update, while error was still decreasing. Fixed-radius refinement was therefore locked before downstream label analysis and applied equally to every condition, without early stopping or validation-dependent updates.

B. Representation and Downstream Development

The representation experiment compared standard and log-standard preprocessing on 8×8, 16×16, 24×24 and 32×32 maps, producing 40 runs with five seeds per condition. Training and validation diagnostics were recorded separately, and no class-performance information informed representation decisions. The shortlist retained log-standard maps of sides eight, 16 and 32 to preserve different capacity trade-offs, while 24×24 maps remained geometric evidence only.

The original downstream grid combined the three shortlisted capacities with $K \in \{10, 15, 20, 30, 40\}$, comparing uniform and training-hit weights under majority mapping or full prevalence correction. Each codebook used five fixed clustering seeds, with k-means++ initialisation and 10 initialisations per fit. The Lloyd algorithm allowed at most 500 iterations with tolerance 10^{-4}, giving 750 clustering fits and 1,500 validation evaluations. Training labels determined the mappings, while validation labels evaluated the resulting classifiers.

Class-score ties selected the lowest class identifier, and unsupported clusters copied the class of the nearest supported centre. Euclidean centre-distance ties selected the lowest cluster identifier, without chaining through unsupported clusters. An unseen observation passed through training-fitted transformation, BMU assignment, codebook-cluster lookup and mapped-class prediction, without neighbouring-unit voting or supervised correction.

After the grid, selection maximised macro F_1 over all 10 classes. Ties preferred higher balanced accuracy, followed by lower SOM variability and a fixed simplicity ordering. Validation therefore provided selection evidence rather than an unbiased estimate of generalisation.

C. Confirmatory Assessment and Baselines

The frozen procedure combined log-standard preprocessing and a 32×32 SOM with $K = 40$, training-hit weighting and $\alpha = 1$. Assessment used five geometry-safe outer folds from the reserved block, with preprocessing and five SOMs fitted on the other four folds. Outer-training observations also supplied clustering weights and class mappings, while development rows were excluded and held-out observations fitted no pipeline stage.

An out-of-fold (OOF) prediction comes from a model fitted without the corresponding held-out fold. Matching SOM and clustering seeds across folds produced 25 OOF vectors, each covering all 206,138 assessment observations exactly once. Metrics used the pooled predictions because averaging fold macro F_1 would give a different nonlinear summary.

Cover and Hart investigated nearest-neighbour classification [3]. The frozen kNN used seven neighbours, Manhattan distance and inverse-distance voting, with votes restricted to zero-distance neighbours whenever present. Numerical features used training median/interquartile-range scaling, and categorical features used one-hot encoding. The classification tree, following the recursive approach developed by Breiman *et al.* [2], used balanced class weights, log-loss splits and maximum depth 16. Minimum split and leaf sizes were five and two, respectively, with pruning parameter 3×10^{-7}. Both Assignment 2 configurations were refitted on the same outer folds using method-specific training-only preprocessing.

D. Post-Hoc Exploratory Sensitivity

Sensitivity analysis was designed after the original assessment had been inspected, prompted by rising development macro F_1 at the cluster-count boundary and severe rare-class false positives under full correction. The later studies examined granularity and correction strength, but the assessment-informed design prevented independent confirmation.

Development exploration reused five frozen log-standard 32×32 codebooks and BMUs, beginning with $K \in \{40, 50, 60, 80, 100, 150\}$ and $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$. A final predeclared extension added $K \in \{200, 250, 300, 400\}$, importing earlier results without refitting. The combined grid comprised 50 conditions, 250 clustering fits and 1,250 evaluations, with the original $K = 40$, $\alpha = 1$ anchor reproduced exactly.

Exploratory ranking used macro F_1, followed by balanced accuracy and lower SOM variability, with remaining ties preferring lower K and α. The first extension exceeded the predeclared 0.03 development-improvement threshold for recommending exploratory assessment, while the final extension retained a hard boundary of 400 clusters. Assessment candidates were frozen at $K = 150$ and $K = 400$, both using $\alpha = 0.5$.

Exploratory assessment reused all 25 fold-specific SOMs and saved BMUs, refitting only clustering and training-label mapping through 250 clustering fits. Identical seed pairs produced complete OOF vectors, while confirmatory and baseline predictions were imported unchanged. No further model development followed exploratory assessment.

E. Aggregation, Diagnostics and Uncertainty

Aggregation averaged five clustering results within each SOM seed, then calculated the mean and sample standard deviation across the five SOM-level means. Mean within-SOM clustering standard deviation described remaining optimisation variation, without treating the 25 seed combinations as independent datasets. Structural diagnostics used labels only after unsupervised fitting and did not influence selection.

Fragmentation counts defined tiny clusters as containing fewer than 10 training observations and singleton clusters as containing exactly a single codebook unit. No cluster was excluded, and the counts remained descriptive without forming another selection score.

Efron and Tibshirani described bootstrap estimation through repeated resampling [4]. The paired bootstrap resampled complete assessment geometry groups with replacement over 2,000 draws, retaining every observation within each sampled group. Identical group multiplicities applied to all methods, with macro F_1 differences averaged across clustering seeds within each SOM seed and then across SOM seeds. Percentile 95% intervals described resampling sensitivity conditional on fixed models and observed groups, without p-values or claims of independent confirmation.

Scientific runs used role-specific seeds, checksum-verified locks and sequential execution with four numerical-library threads. The Linux implementation and replayable records were available in the repository.[1]

V. Results

The results present representation quality, original development selection and confirmatory assessment before the separate post-hoc sensitivity and transfer findings.

A. Representation Quality and Capacity

Table II summarises the 70-epoch experiment, reporting validation CQE with sample standard deviations alongside validation TE and training-BMU occupancy. Figure 1 shows the capacity trends with five-seed standard-deviation error bars. Log-standard preprocessing gave lower validation CQE at every capacity, reaching 0.1412 on the largest map compared with 0.1665 under standard preprocessing. Lower contextual distance indicated closer codebook fit within the transformed spaces, without establishing a transformation-independent advantage.

TABLE II
Development representation diagnostics

Representation	Map	Validation CQE	TE	Occupancy (%)
Standard	8×8	0.4116 ± 0.0080	0.114	92.8
Standard	16×16	0.2548 ± 0.0019	0.151	87.2
Standard	24×24	0.1981 ± 0.0021	0.150	82.7
Standard	32×32	0.1665 ± 0.0019	0.145	80.4
Log-standard	8×8	0.3130 ± 0.0019	0.081	93.4
Log-standard	16×16	0.2057 ± 0.0009	0.111	87.7
Log-standard	24×24	0.1625 ± 0.0012	0.129	85.8
Log-standard	32×32	0.1412 ± 0.0005	0.154	84.6

Increasing capacity reduced CQE with diminishing returns, as the log-standard improvement from 8×8 to 16×16 exceeded the improvement from 24×24 to 32×32. Occupancy fell and validation TE increased, while the largest standard map preserved lower TE than the largest log-standard map. Log-standard preprocessing therefore improved contextual fit without improving every diagnostic, and topology varied more across seeds than CQE. For the log-standard 8×8 map, TE had standard deviation 0.042 around a mean of 0.081.

Fixed-radius refinement flattened the convergence histories, with mean validation QE changes below 0.03% in magnitude during the last 10 updates for every log-standard capacity. Final mean movement on the largest log-standard map was approximately 0.0009 transformed Euclidean units, supporting practical stabilisation without establishing an exact fixed point.

Figure 2 presents an unlabelled development map selected by median original development performance within the seed hierarchy. The neighbour-distance panel averages distances to edge-sharing codebook vectors at each unit, while training hits use a logarithmic scale. Duration and source bytes were predeclared semantic features, shown in transformed coordinates centred around zero. Together, the panels illustrate unequal support and spatial variation without class overlays.

B. Original Development Selection

The original grid selected log-standard 32×32 geometry with 40 hit-weighted clusters and full correction, achieving validation macro F_1 of 0.3164 with SOM standard deviation 0.0049. Intermediate and compact maps achieved 0.3066 and 0.2979, respectively, so macro F_1 selection favoured the largest map despite higher balanced accuracy for the intermediate map.

At the selected capacity and cluster count, majority mapping achieved macro F_1 of 0.2654 and accuracy of 0.6932. Full correction raised macro F_1 to 0.3164 but lowered accuracy to 0.5851, while mean predicted-class coverage increased from 4.4 to 9.92. Corrected Worms recall reached approximately 0.714, yet 387 false positives accompanied only seven true validation cases. Broader class coverage therefore did not establish adequate precision.

Under otherwise matching corrected settings, uniform weighting achieved macro F_1 of 0.3026, below the training-hit-weighted result. Performance was still rising at 40 clusters, but the procedure was frozen at the declared boundary before confirmatory assessment.

C. Confirmatory Assessment

The 40-cluster procedure achieved pooled assessment macro F_1 of 0.324, with SOM-level standard deviation 0.0060 and mean within-SOM clustering standard deviation 0.0078. The estimates remained the primary confirmatory results, compared with same-fold macro F_1 of 0.523 for kNN and 0.583 for the tree. The tree achieved the highest balanced accuracy, whereas kNN achieved the highest accuracy and weighted F_1.

Across 25 fold-specific SOMs, mean training and held-out CQE remained close at approximately 0.1365 and 0.1384, with corresponding TE of 0.1256 and 0.1337. Comparable diagnostics supported geometric transfer to unseen groups despite weak classification performance. Every cluster had training support, ruling out unsupported-cluster fallback as a source of classification errors.

Rare-class recall nevertheless accompanied extensive false positives, particularly for the 139 assessment Worms observations. Worms recall averaged 0.707 but precision only 0.013,

[1] github.com/adriaandh1/ml741-assignmen4-27284808

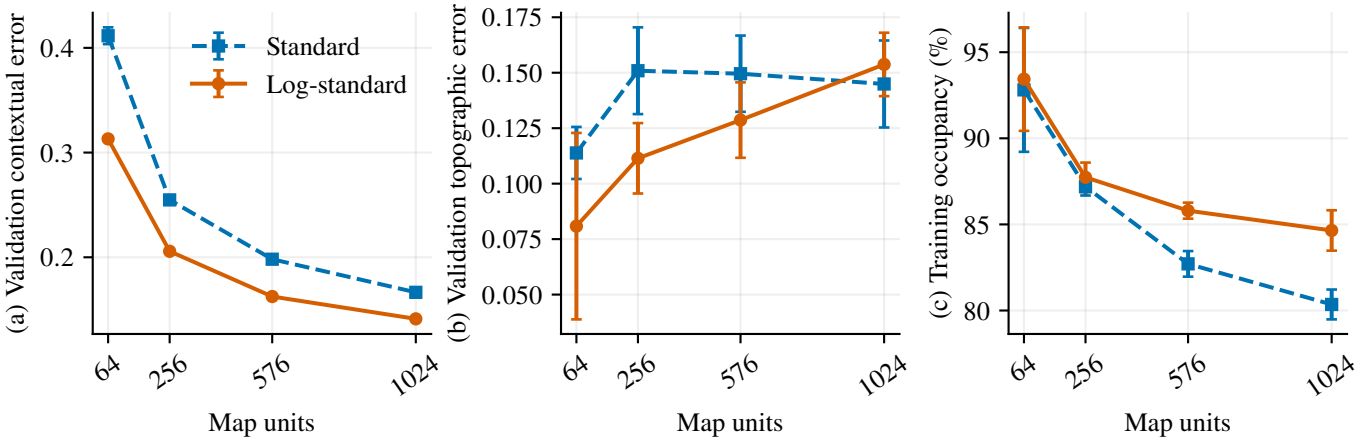


Fig. 1. Development representation diagnostics across map capacity

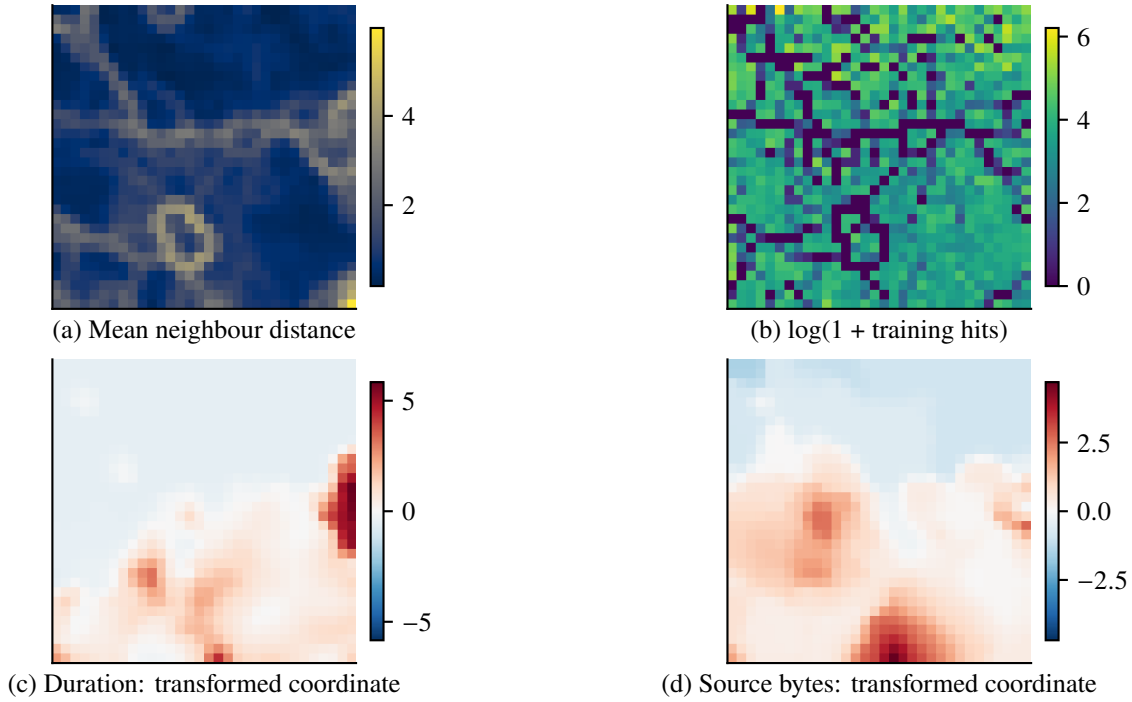


Fig. 2. Unlabelled development geometry and component planes

with approximately 98 true positives and 7,205 false positives per replicate. Shellcode showed a similar imbalance, with recall 0.540 and precision 0.030, motivating the subsequent sensitivity study.

D. Post-Hoc Development Sensitivity

Figure 3 shows all 50 nested means from varying granularity and correction on unchanged SOMs. A dashed boundary separates the first extension from the final higher-granularity extension. Macro F_1 increased with cluster count at every tested correction strength, and partial correction gave the highest mean at every count. The best α was 0.75 at 60, 80 and 100 clusters, and 0.5 elsewhere.

At $\alpha = 0.5$, the 150-cluster procedure achieved development macro F_1 of 0.4196, a gain of 0.1032 over the frozen procedure. The final extension reached 0.4475 at 400 clusters, although gains at fixed $\alpha = 0.5$ fell from 0.0284 for 100–150 clusters to 0.0026 for 300–400 clusters. The diminishing gains did not establish a plateau or global optimum.

Figure 4 relates classification to fragmentation at fixed $\alpha = 0.5$, with SOM-level standard-deviation error bars. Singleton fractions count clusters containing a single unit, while tiny fractions count clusters containing fewer than 10 training observations. Silhouette fell as singleton clusters increased, reaching singleton and tiny fractions of approximately 27.5% and 2.5% at 400 clusters. All clusters retained positive support, so the predictive gains concerned finer partitions of the same

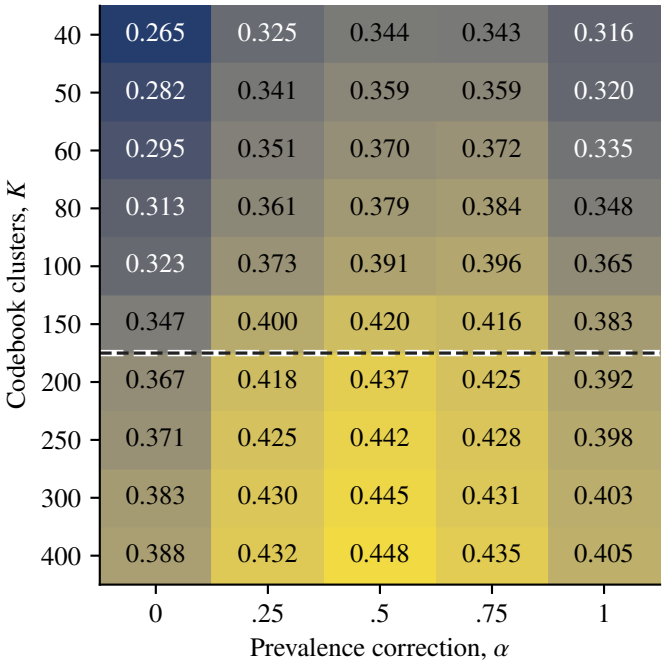


Fig. 3. Post-hoc development sensitivity to granularity and correction

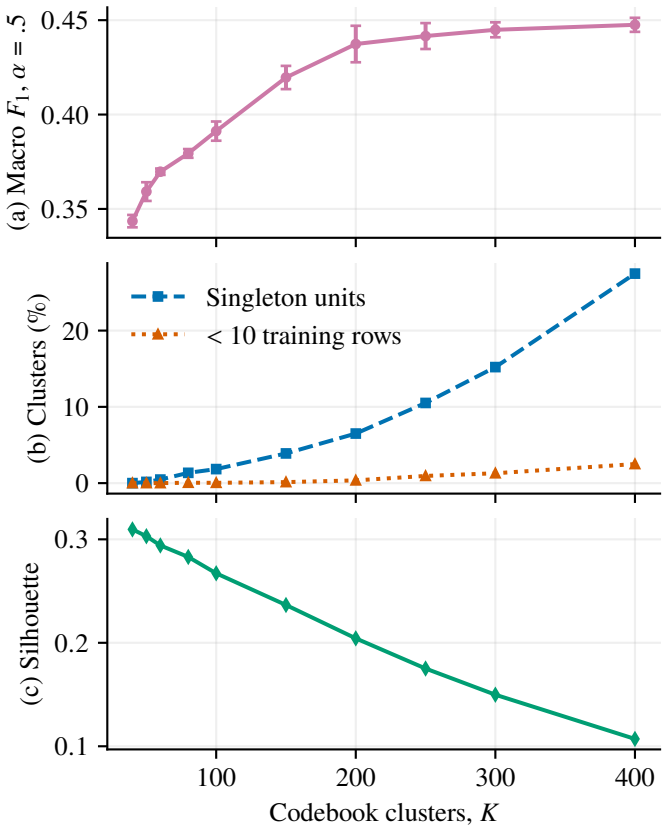


Fig. 4. Post-hoc development performance and codebook fragmentation

codebook without changing SOM quantisation.

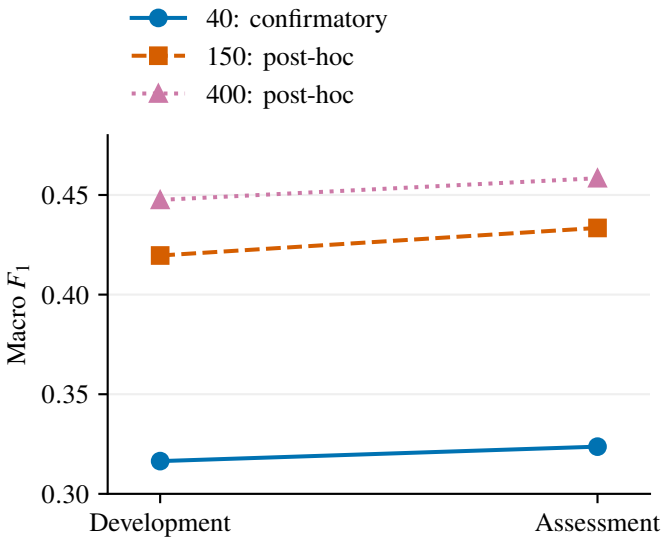


Fig. 5. Development and assessment macro F_1 for the map-derived procedures

E. Post-Hoc Assessment Transfer and Class Effects

Table III compares all five procedures on shared folds, with macro F_1 uncertainty denoting sample standard deviation across five SOM-level means. Figure 5 compares development and assessment performance, distinguishing the confirmatory 40-cluster result from the post-hoc larger-cluster results. Development gains transferred to the already-viewed assessment block, but the transfer did not provide independent confirmation.

At fixed $\alpha = 0.5$, the 400-cluster procedure exceeded the 150-cluster procedure by 0.0250 assessment macro F_1, with positive differences across all 25 paired stochastic replicates. The paired group-bootstrap interval was [0.0233, 0.0266], while the gain over the confirmatory procedure was 0.1347 with interval [0.1308, 0.1384]. Both intervals described resampling sensitivity under the post-hoc procedures.

Figure 6 retains all 10 classes, showing improvements for Reconnaissance and DoS relative to the coarse, fully corrected procedure. Generic maintained high class F_1, while finer clustering improved Shellcode without reaching either supervised baseline. Backdoor and Analysis continued to have low class F_1.

Table IV reports precision, recall and false-positive counts for four rare classes, with fractional counts obtained by averaging 25 integer confusion matrices. Finer clustering with partial correction sharply reduced Worms false positives but also reduced recall, giving approximately 180 false positives and recall 0.072 at 150 clusters. Increasing to 400 clusters raised recall to 0.127 alongside approximately 451 false positives, lowering precision and leaving Worms F_1 near 0.059 under both procedures. Weak recall across 139 assessment cases therefore extended beyond the uncertainty of the seven-case development sample.

Assessment-fold codebooks reproduced the fragmentation trend, with singleton clusters increasing from 4.46% at $K = 150$ to 27.53% at $K = 400$ as mean silhouette fell from 0.241 to 0.108. Held-out purity rose from 0.754 to 0.768, but NMI

TABLE III
ASSESSMENT PERFORMANCE ON THE SHARED OUTER FOLDS

Procedure	Status	Macro F_1	Balanced accuracy	Accuracy	Weighted F_1
SOM K40	Confirmatory	0.3237 ± 0.0060	0.4611	0.5868	0.6455
SOM K150	Post-hoc	0.4335 ± 0.0021	0.4569	0.7178	0.7352
SOM K400	Post-hoc	0.4584 ± 0.0017	0.4915	0.7307	0.7469
kNN	Frozen baseline	0.5233	0.5060	0.8008	0.7957
Tree	Frozen baseline	0.5834	0.6661	0.7566	0.7821

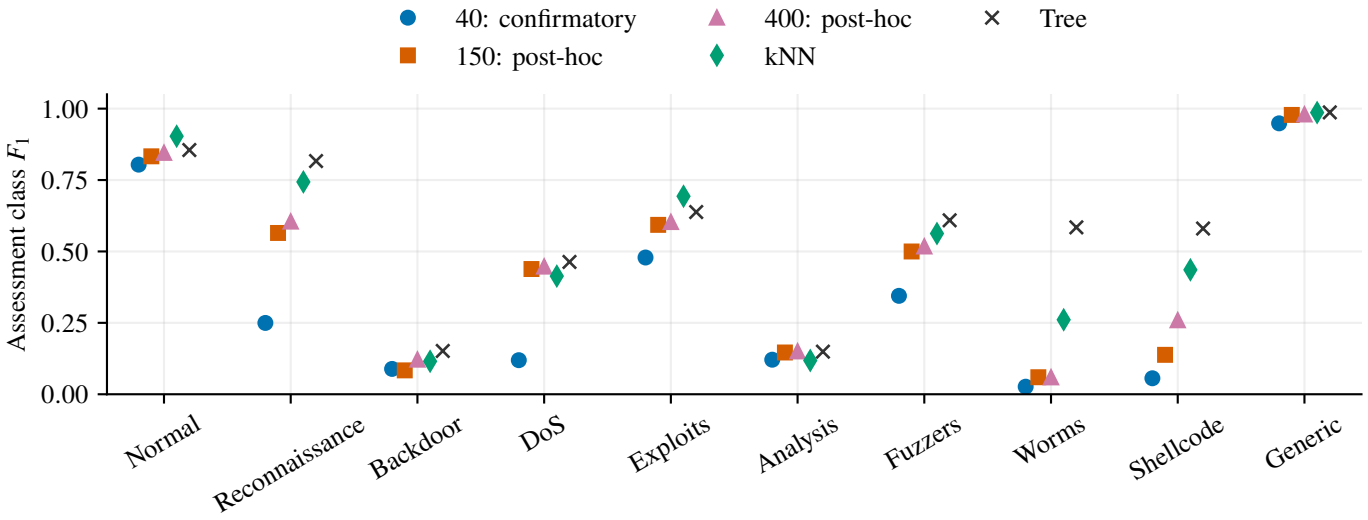


Fig. 6. Class-specific assessment F_1 on the shared outer folds

TABLE IV
RARE-CLASS ASSESSMENT PERFORMANCE OF THE MAP-DERIVED PROCEDURES

Class (true support)	K / α	Precision	Recall	F_1	True positives	False positives	False negatives
Backdoor (1,864)	40 / 1	0.049	0.511	0.089	952.2	18,699.8	911.8
Backdoor (1,864)	150 / .5	0.115	0.070	0.084	130.1	1,009.9	1733.9
Backdoor (1,864)	400 / .5	0.125	0.118	0.121	220.8	1,557.9	1643.2
Analysis (2,141)	40 / 1	0.107	0.152	0.121	325.0	2,999.9	1816.0
Analysis (2,141)	150 / .5	0.210	0.112	0.146	240.6	909.8	1900.4
Analysis (2,141)	400 / .5	0.221	0.113	0.150	242.8	857.6	1898.2
Worms (139)	40 / 1	0.013	0.707	0.026	98.3	7,205.3	40.7
Worms (139)	150 / .5	0.058	0.072	0.060	10.0	179.6	129.0
Worms (139)	400 / .5	0.039	0.127	0.059	17.6	451.4	121.4
Shellcode (1,208)	40 / 1	0.030	0.540	0.056	652.8	21,552.8	555.2
Shellcode (1,208)	150 / .5	0.147	0.137	0.138	164.9	981.6	1043.1
Shellcode (1,208)	400 / .5	0.220	0.315	0.259	380.4	1,353.4	827.6

and ARI fell, showing that improved prediction did not imply improvement under every structural measure.

The 400-cluster procedure remained below both supervised baselines, with a macro F_1 gap to kNN of −0.0649 and paired interval [−0.0761, −0.0527]. The tree gap was −0.1250, with interval [−0.1350, −0.1147], retaining the descriptive, post-hoc interpretation.

VI. Discussion

Useful feature organisation did not necessarily translate into reliable attack prediction. The discussion considers representation, mapping and class-specific effects before the supervised comparison and study limitations.

A. Representation and Predictive Utility

For the first research question, larger maps reduced contextual distance but also lowered occupancy and weakened local adjacency preservation, exposing a capacity trade-off. Similar training and held-out diagnostics supported geometric transfer within the supplied extract, although geometric transfer alone did not establish class separation.

For the second research question, the SOM and codebook clustering described feature structure, while training-count mapping made the complete classifier partly supervised. Conflicting targets for identical inputs also constrained deterministic separation independently of clustering, allowing useful geometric organisation to coexist with poor minority-class precision.

At fixed correction strength, finer codebook partitions improved class prediction while producing more singleton clusters and lower silhouette. The opposing trends reflected different objectives, since silhouette measured compactness and separation whereas macro F_1 evaluated mapped-class decisions. The experiment did not establish fragmentation as the cause of the predictive gain.

Purity favoured subdivision on assessment-fold models, but falling NMI and ARI showed that structural measures did not provide interchangeable judgements of predictive utility. Comparing 150 and 400 clusters at fixed $\alpha = 0.5$ isolated granularity more closely than comparing 40 and 400 clusters, where both granularity and correction strength changed.

B. Imbalance and Class-Specific Effects

Full correction divided local class counts by global training frequencies, giving rare classes a greater relative advantage than partial correction. Weak local counts for rare classes consequently competed with much larger common-class counts, consistent with the observed thousands of rare-class false positives. Partial correction reduced the advantage, but the reduction also sacrificed recall.

Rarity alone did not account for the class outcomes, as Shellcode benefited from finer partitions while Worms retained weak precision and recall. Backdoor remained difficult, Analysis changed modestly and Generic retained high class F_1. The seven-case development Worms sample introduced uncertainty, but weak recall persisted on the larger assessment sample.

For the third research question, the classification tree achieved the highest macro F_1 and balanced accuracy, whereas kNN achieved the highest accuracy and weighted F_1. Training labels directly informed tree boundaries and neighbour votes, while SOM geometry remained label-free. However, preprocessing and tuning also differed between methods, preventing the comparison from isolating representation learning from the remaining pipeline.

For the fourth research question, saved representations allowed inexpensive downstream sensitivity analysis. Training 25 assessment SOMs took approximately 103 minutes, compared with approximately 167 seconds for exploratory assessment, including analysis and bootstrap. Reusing the maps avoided further SOM, preprocessing, BMU or baseline fitting.

C. Limitations and Experimental Status

The post-hoc studies followed original assessment inspection and repeatedly reused validation, so later candidate freezing did not remove the assessment-informed research design. Bootstrap intervals described resampling of observed geometry groups conditional on fixed procedures, excluding selection uncertainty and transfer to another acquisition environment.

The exploratory search ended at the predeclared 400-cluster boundary without establishing either a plateau or global optimality, and no further development followed exploratory assessment. Seed variation measured initialisation sensitivity on fixed partitions rather than replication on independent data.

Geometry-group isolation prevented identical inputs from crossing folds but did not provide temporal, host or deployment separation, leaving possible dependencies from near-duplicate traffic and unrecorded acquisition structure. Unknown categories also shared a zero block, while training-only vocabularies produced 190–192 assessment coordinates compared with 150 during development. Different vocabularies consequently limited direct comparison of raw QE between the settings.

No assessment prediction or performance informed confirmatory selection, although an inherited integrity test parsed full-source labels before repair of the strict gate. The parsing produced no assessment metric or selection effect, distinguishing source-label reading from performance-guided selection.

The conclusions remained specific to the supplied extract and taxonomy, and higher macro F_1 did not establish acceptable false-positive rates for deployment.

VII. Conclusion

The self-organising map (SOM) produced a practically stabilised traffic representation with comparable training and held-out geometry, while log-standard preprocessing reduced contextual distance. Larger maps improved codebook fit but reduced occupancy and weakened local topology preservation, so geometric quality alone did not ensure reliable attack-class prediction.

The confirmatory classifier achieved macro F_1 of 0.324, with full prevalence correction increasing rare-class recall at a severe false-positive cost. Post-hoc finer clustering with partial correction raised assessment macro F_1 to 0.458 at 400 clusters, accompanied by greater fragmentation and lower silhouette. Worms recognition remained weak, and the upper search boundary left the granularity optimum unresolved.

The k-nearest-neighbour (kNN) and classification-tree baselines retained higher overall macro F_1 on shared folds, while the exploratory estimate remained separate from the primary confirmatory result. Useful unsupervised geometry did not guarantee dependable class separation, while mapping exposed the precision–recall cost of prevalence correction.

Appendix A
Acronyms

ARI	adjusted Rand index
BMU	best matching unit
CQE	contextual quantisation error
DoS	denial of service
kNN	k-nearest neighbour
NMI	normalised mutual information
OOF	out-of-fold
QE	quantisation error
SOM	self-organising map
TE	topographic error
UNSW-NB15	University of New South Wales Network Based 2015

Appendix B
Symbols

$\mathbf{a}_k^{(e)}$	sum of training vectors assigned to source unit k in epoch e
α	prevalence-correction strength
b	best-matching-unit index function
c	target-class index
d	contextual quantisation-error function
e	batch-update epoch index
f_c	false positives for class c
$F_{1,c}$	precision–recall score for class c
$F_{1,\text{macro}}$	mean class score across all 10 classes
g	number of target classes
$h_{jk}^{(e)}$	Gaussian influence from source k to destination j in epoch e
j	destination map-unit index
k	source map-unit index
K	number of codebook clusters
ℓ	map side length
m	number of map units
$\boldsymbol{\mu}_T$	training feature centroid
n_c	total training count of class c
$n_k^{(e)}$	training count assigned to source unit k in epoch e
n_X	number of evaluated observations in X
n_{zc}	training count assigned to cluster z and class c
p_c	precision for class c
q	quantisation-error function
r_c	recall for class c
$\mathbf{r}_j$	lattice coordinate of unit j
s_{zc}	prevalence-correction score function for cluster z and class c
σ_e	neighbourhood radius in epoch e
t_c	true positives for class c
T	training partition
u_c	false negatives for class c
v	non-negative numerical feature value
$\mathbf{w}_j$	codebook vector of unit j
$\mathbf{x}$	transformed observation vector
X	evaluated partition
z	codebook-cluster index

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